MIL-C-60644(MU)
1 February 1968
SUPERSEDING
FA-PD-MI-2555, Rev. 0
9 July 1962

MILITARY SPECIFICATION
CARTRIDGE, DELAY: M175 ASSEMBLY

1. SCOPE

1.1 This specification covers Cartridge, Delay: M175 Assembly, for use with propellant actuated devices in Aircrew emergency escape systems.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Military

MIL-A-2550	- Ammunition and Special Weapons, General Specifications for
MIL-E-5272	- Environmental Testing, Aeronautical and Associated Equipment, General Specification for
MIL-C-60539	- Anodic Coating, Conventional, For Aluminum and Aluminum Alloys

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-413	- Visual Inspection Guide for Rubber O-Rings
MIL-STD-639	- Surface Roughness Values Preferred for Army Ordnance Materiel
MIL-STD-1167	- Ammunition Data Card
MS-28741	- Hose Assembly - Detachable End Fitting, Medium Pressure

FSC 1377

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2.1 (Cont'd)

DRAWINGS

Munitions Command	
D10522660	- Cartridge, Delay: M175 Assembly
C8864168	- Carton, Inner Packing, Ammunition for Cartridge, Delay: M175
C8864169	- Carton, Outer Packing, Ammunition for Cartridge, Delay: M175
IEL-10522660	- Inspection Equipment List for Cartridge, Delay: M175 Assembly

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procurement activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 General.- Materials, parts and assemblies shall conform to the applicable drawings and referenced specifications.

3.2 Protective finish.- Protective finish shall be in accordance with MIL-C-60539.

3.3 Vibration.- The assembled cartridge shall withstand vibration as prescribed in MIL-E-5272, Procedure XII without evidence of any disintegration of the delay charge, primer, igniter or propellant, or derangement of components.

3.4 Carbon monoxide gas.- The cartridge, when assembled into an accepted, unused M89 Initiator, temperature conditioned at -65°F. to -70°F., 65°F. to 75°F. and 160°F. to 165°F. and fired into a sealed total volume of 0.70 cubic inches shall not produce carbon monoxide gas leakage greater than 20.0 cubic centimeters total. Volume of gas leakage shall be recorded at 10 minutes and 60 minutes after firing.

3.5 Functional.

3.5.1 The cartridge shall not fail to fire.

3.5.2 Ballistics.- The cartridge when tested in an M89 initiator shall comply with the requirements prescribed in Table I.

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3.5.2 (Cont'd)

TABLE I

Temperature	Pressure (PSI) 1/ Min.	Max.	Delay Seconds
-65° to -70°F	500	1500	6.0 Max.
65° to 75°F	575	1500	--
160° to 165°F	625	1500	4.0 Min.

1/ Pressure values shall be produced in a one cubic inch pressure station at the end of 15 feet of hose (Size No. 4) in accordance with MS-28741.

3.6 Workmanship.- Workmanship shall be in accordance with the requirements of the applicable drawings, referenced specifications and the following:

3.6.1 Metal defects.- The metal part shall be free from cracks, splits, cold shuts, inclusions, porosity or other metal defects.

3.6.2 Burr.- The part shall not have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

3.6.3 Foreign matter.- The part or assembly shall not contain chips, dirt, grease, oil, rust, corrosion or other foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies conform to prescribed requirements.

4.1.1 Quality Assurance terms and definitions.- Reference shall be made to MIL-STD-109 to define the quality assurance terms used.

4.1.2 Inspection.- Inspection shall be in accordance with MIL-A-2550.

4.2 First article sample.

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4.2.1 Initial production sample.- At the beginning of regular production, a sample shall be submitted in accordance with contract requirements and shall consist of 23 anodized units for protective finish tests, 50 units for hardness tests, 50 for carbon monoxide tests, 12 for vibration (to be selected from ballistics sample), 5 for sealant tests and 30 for ballistic tests and tested in accordance with Table II and 10 sets of unassembled components (including explosive components) for dimensional and visual examination.

TABLE II

Paragraph Requirement	Test Procedure	Test	Sample Size	Type of Test Destr.	Non-Destr.
3.2	4.4.1	Protective finish	23	X	-
Dwg.	4.4.2	Hardness	50	-	X
Dwg.	4.4.3	Airtightness	Each	-	X
3.3	4.4.4	Vibration	12 from ballistics	-	X
3.4	4.4.5	Carbon Monoxide	50	X	-
Dwg.	4.4.6	Sealant application	5	X	-
3.5.2	4.4.7	Ballistics	30	X	-

The sample shall be manufactured using the same materials, equipment, processes and procedures as used in regular production. All parts and materials, including packaging and packing shall be the same as used for regular production and shall be obtained from the same source of supply.

4.2.1.1 Examination and tests.- After examination and provisional acceptance at source, the sample shall be inspected for all requirements of the drawings and specifications at a Government laboratory or such other facility specified in the contract.

4.2.1.2 Ballistics.- The sample for ballistic tests shall be temperature conditioned and shall conform to requirements as indicated in Tables I and III.

TABLE III

Temperature	Vibrated Cartridges	Not Vibrated
-65° to -70°F	4	6
65° to 75°F	4	6
160° to 165°F	4	6

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4.2.1.3 Initial production sample failure.- Failure of the sample to comply with the requirements of the drawings and specifications shall result in sample disapproval.

4.3 Inspection provisions.

4.3.1 Lot.

4.3.1.1 Submission of product.- The product shall be submitted in accordance with MIL-STD-105.

4.3.1.2 Size of lot.- Unless otherwise specified by the procuring agency, a lot shall consist of cartridges made up of:

(a) Components produced by a homogeneous process

(b) Propellant from not more than one lot

(c) Primers from not more than one lot

(d) Igniters from not more than one lot.

4.3.2 Examination.- One hundred percent examination shall be performed for all critical defects. Examination for major and minor defects shall be performed on a defect basis in accordance with the classification of defects contained herein, using the sampling plan and associated criteria in Tables IV and V. For an individual defect (single characteristic), the criteria in Table IV shall be used. For total defects within a major or minor class, the criteria in Table V shall be used. All non-conforming material shall be rejected. The Ordnance Standard Finish Blocks prescribed in MIL-STD-639 shall be used as a basis of comparison for surface roughness determination. Inspection for rubber O-rings shall be in accordance with MIL-STD-413. Four power magnification (4X) may be used in performing visual examination for such defects as burr and foreign matter.

TABLE IV

Lot Size	Sample Size	Major Accept	Major Reject	Minor Accept	Minor Reject
1 - 300	100% Examination	-	-	-	-
301 - 1300	110	1	2	2	3
1301 - 3200	150	2	3	3	4
3201 - 8000	225	3	4	4	5
8001 - 22000	300	4	5	5	6

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4.3.2 (Cont'd)

TABLE V

Sample Size for Major Defects	Total Allowable Defects Per Sample Size			
	110	150	225	300
Number of Defects in Class				
2 - 3	1	2	3	4
4 - 7	2	3	4	5
8 - 19	3	4	5	7
20 - 43	4	5	8	10
Over 43	6	8	11	14
Sample size for Minor Defects	110	150	225	300
Number of Defects in Class				
2 - 3	2	3	4	5
4 - 7	3	4	5	7
8 - 16	4	5	8	10
17 - 48	6	8	11	14
Over 48	8	11	17	20

4.3.2.1 Delay Element, XM45, Drawing D10522658

Defect	Method of Inspection
CRITICAL	
1. Metal defective	Visual
2. Mixed ammunition	Visual
3. Flash hole missing or obstructed	Visual
4. Marking missing, incorrect or illegible	Visual
5. Delay charge not visible through flash hole	Visual
MAJOR	
101. Protective finish inadequate or defective	Visual
102. Foreign matter	Visual
MINOR	
None defined	

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4.3.2.2 Ring, Retaining, Drawing 8597235 covering a detail of Drawing D10522660

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
101. Concentricity of inside diameter with outside diameter	Gage
102. Diameter inside, max.	Gage
103. Diameter outside, min.	Gage
104. Thickness	Gage
105. Metal defective	Visual
106. Protective finish inadequate or defective	Visual
107. Burr	Visual
108. Foreign matter	Visual
MINOR	
201. Diameter inside, min.	Gage
202. Diameter outside, max.	Gage
203. Surface finish improper	Visual

4.3.2.3 Case, Cartridge, Drawing 10519996, covering a detail of Drawing D10522660

CRITICAL	
1. Metal defective	Visual
MAJOR	
101. Concentricity of head with body	Gage
102. Diameter of head, min.	Gage
103. Thickness of wall, min.	Gage
104. Thickness of base, min.	Gage
105. Protective finish inadequate or defective	Visual
106. Burr or sharp corner	Visual
107. Foreign matter	Visual

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4.3.2.3 (Cont'd)

	Defect	Method of Inspection
MINOR		
201.	Diameter of head, max.	Gage
202.	Diameter of body	Gage
203.	Total length	Gage
204.	Thickness of wall, max.	Gage
205.	Thickness of base, max.	Gage
206.	Location of datum diameter	Gage
207.	Radius missing or incorrect	Visual
208.	Surface finish improper	Visual

4.3.2.4 Assembly, Cartridge Delay, M175 Assembly, Drawing D10522660

CRITICAL		
1.	Any part missing	X-Ray 1/ and Visual
2.	Improper assembly	X-Ray 1/ and Visual
3.	Metal defective	Visual
4.	Marking missing, incorrect or illegible	Visual
MAJOR		
101.	Concentricity of head with body	Gage
102.	Diameter of crimp, min.	Gage
103.	Location of primer head	Gage
MINOR		
201.	Diameter of body, max.	Gage
202.	Diameter of head, max.	Gage
203.	Total length, max.	Gage
204.	Location of datum diameter	Gage
205.	Varnish missing	Visual

1/ The cartridge shall be x-rayed with the long axis of the cartridge perpendicular to the rays of the x-ray machine. The x-ray negatives, properly identified, shall be forwarded to the procuring agency.

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4.3.2.5 Carton, Inner Packing, Ammunition for Cartridge, Delay: M175, Drawing C8864168

Defect	Method of Inspection
CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Contents loose	Manual and Visual
202. Quantity of contents incorrect	Visual
203. Separator or filler missing	Visual
204. Heat sealer incomplete, loose or distorted	Visual
205. Cut or damage through impregnated box	Visual
206. Marking missing, incorrect or illegible	Visual

4.3.2.6 Carton, Outer Packing, Ammunition for Cartridge, Delay: M175, Drawing C8864169

CRITICAL	
None defined	
MAJOR	
None defined	
MINOR	
201. Contents loose	Manual and Visual
202. Hardware missing, loose, broken or improperly engaged	Visual
203. Strapping missing, loose, broken or improperly assembled	Visual
204. Contents of box exposed	Visual
205. Box damaged	Visual
206. Marking missing, incorrect or illegible	Visual
207. Number of cartons incorrect	Visual
208. Car seal missing, unsealed or improperly positioned	Visual

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4.3.3 Inspection testing.- The following tests shall be performed in accordance with the provisions of 4.4.

4.3.3.1 Protective finish.- Twenty three units shall be selected from each anodized lot for protective finish tests. Failure of any unit to comply with the requirements of 3.2 shall be cause for rejection of the lot.

4.3.3.2 Hardness.- Fifty units shall be selected for this test. Failure of any unit to comply with the drawing requirements shall be cause for rejection of the lot.

4.3.3.3 Airtightness.- Each unit shall be subjected to the airtightness test. Failure of any unit to comply with the drawing requirements shall be cause for rejection of the unit.

4.3.3.4 Vibration.- Twelve of the 30 assembled x-rayed cartridges of the initial production sample selected for ballistics tests and properly identified shall be subjected to vibration as in 3.3. After the vibration test, the 12 vibrated cartridges shall be x-rayed as directed in Note 1/, 4.3.2.4. Evidence of any disintegration of the primer, propellant or igniter or derangement of any components shall be cause for rejection of the lot as indicated in 3.3. When a change occurs in the lot of primers, propellant or igniters after the initiation of production, 12 assembled cartridges from the first production lot shall be tested in accordance with this paragraph.

4.3.3.5 Carbon monoxide gas leakage.- Fifty assembled cartridges from the initial production sample and 25 cartridges from each subsequent lot shall be tested for carbon monoxide gas leakage. Failure of any cartridge to comply with the requirements of 3.4 shall be cause for rejection of the lot.

4.3.3.6 Sealant application.- Five cartridges from each production lot shall be utilized for this test. Failure of any cartridge to comply with the drawing requirements shall be cause for rejection of the lot.

4.3.3.7 Functional.- The following tests shall be performed by the Government.

4.3.3.7.1 Ballistics.- A sample of 30 assembled cartridges from each production lot shall be ballistically tested after temperature conditioning in accordance with Table VI and 4.4.6.1.

TABLE VI

Temperature Degrees (°) Fahrenheit (F)	Sample
-65° to -70°	10
65° to 75°	10
160° to 165°	10

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4.3.4 Inspection equipment.- The examination and tests shall be performed using equipment listed on IEL-10522660.

4.4 Test methods and procedures.

4.4.1 Protective finish.- Test procedures for protective finish shall be in accordance with MIL-C-60539.

4.4.2 Hardness.- The hardness test shall be performed on all heat-treated components prior to the application of protective finish. A hardness testing machine shall be used to determine compliance with the drawing requirements.

4.4.3 Airtightness.- The method for determining airtightness shall be by use of the equipment listed on IEL-10522660.

4.4.4 Vibration.- The method and procedure for performing the vibration test shall be in accordance with MIL-E-5272, Procedure XII.

4.4.5 Carbon monoxide gas leakage.- The test shall be performed utilizing the equipment listed on IEL-10522660.

4.4.6 Sealant application.- The cartridges shall be disassembled in such a manner as to expose the firing plug and primer area for examination.

4.4.7 Ballistics.- The test sample shall be rigidly supported in the fixture. The sample shall be tested within 5 minutes after removal from the conditioning chamber. When this time limit is exceeded, the sample shall be reconditioned in accordance with 4.4.7.1.

4.4.7.1 Temperature conditioning.- Cartridges shall be loaded into acceptable M89 initiators with end caps to cover the gas ports during the conditioning period. The temperature of the propellant within the cartridge shall be considered as the conditioning temperature and will be determined by means of a thermocoupler attached to the center of the propellant charge of a specimen cartridge assembled in a standard test device similar to the initiators undergoing tests. The conditiong time shall be a minimum of one hour longer than the time necessary to bring the element contained in the specimen cartridge to the required temperature.

5. PREPARATION FOR DELIVERY

5.1 Packing and marking.- Packing and marking shall be in accordance with Drawings C8864168 and C8864169, except that all markings shall refer to Cartridge, Delay: M175.

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5.2 Data card.- Data card information shall be in accordance with MIL-STD-1167.

6. NOTES

6.1 Ignition delay is defined as the time that elapses between firing of the primer and the beginning of a continuous pressure rise on a pressure time record.

Custodian:

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Preparing activity:

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

[ ] YES [ ] NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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